

Shervin Darmanki Farahani

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EDUCATION

University of Toronto

Toronto, ON

Bachelor of Applied Science in Computer Engineering + Co-op, Minor in AI Engineering

Expected 2028

- Relevant Courses: Programming Fundamentals, Algorithms and Data Structures, Operating Systems, Digital Systems, Computer Organization, Software Design and Communication, Signals and Systems

Bachelor of Applied Science in Engineering Science, Dean's List Scholar

Sep. 2023 – May 2024

- Relevant Courses: Introduction to Computer Programming, Computer Algorithms and Data Structures

EXPERIENCE

Software Engineer Intern

May 2026 – Present

Advanced Micro Devices (AMD)

Markham, ON

- Resolved **50+ production defects** in **Radeon Software** using **C++**, spanning **QML** UI behavior, feature logic, **hardware detection and interaction**, and data ordering across a large customer-facing codebase.
- Performed **root-cause analysis** on **10+ search, feature, and hardware-behavior issues**, tracing interactions across **interconnected components** and implementing fixes beyond localized code changes.
- Collaborated with **senior and full-time engineers** on **3+ cross-component fixes**, navigating unfamiliar code, requesting technical guidance, and **restructuring existing logic** to satisfy complex feature requirements.

Software Engineer / Cloud Infrastructure & DevOps Intern

Jan 2026 – May 2026

Royal Bank of Canada (RBC)

Toronto, ON

- Developed a **malware scanning pipeline** in **Python** to process vendor file uploads, integrating with a scanning API to automatically **quarantine malicious files** and securely route clean files to **AWS S3 buckets**, improving system security and reducing manual inspection overhead by **90%+**.
- Engineered a **data validation system** that ingests **CSV files from S3**, populates databases, and executes **SQL-based consistency checks** against production data, enabling early detection of discrepancies.
- **Provisioned VM's** and configured **custom CI/CD runners**, enabling custom deployment to **WPVIP**.
- Collaborated on **production deployments** using **Java** and **JavaScript**, ensuring reliable feature releases.

PROJECTS

CHIP-8 Emulator | C++, WxWidgets, OpenGL, SDL

September 2025 - Present

- Developed a **CHIP-8 emulator in C++** with **35+ opcodes**, achieving 100% compatibility with all ROMs.
- Implemented an **OpenGL graphics pipeline** at 60 FPS, with accurate input handling via keyboard mappings.
- Integrated **SDL-based audio system** to emulate **hardware timers** and sound effects, achieving $\leq 5\text{ms}$ latency.
- Designed and built a **wxWidgets-powered desktop GUI** with ROM loading, adjustable emulation speeds, pause/resume, and reset features, boosting usability efficiency by 40%.

OpenStreetMap Mapper Software (ECE297 Project) | C++, CSS, Git

January 2025 – May 2025

- Designed and implemented an **efficient API** for querying and processing OSM street data, achieving $\mathcal{O}(1)$ average case look-ups through optimized data structures (e.g., **hash maps, tries**)
- Integrated **GTK and EZGL** libraries to render OSM maps with real-time zoom, pan, and search functionality, supporting up to **10,000+ map elements** with $\leq 100\text{ms}$ rendering latency
- Solved a traveling salesman-style problem by implementing a **multi-target Dijkstra's Algorithm** as well as Simulated Annealing and 2/3-opt moves, **reducing travel time by 16%**.

Systems Simulation Projects (ECE244, ECE243, ECE344 Projects) | C, C++

September 2024 - Present

- Developed 5+ **C++ applications** from scratch, applying **OOP** and **data structures**; projects included a **Pong game** with **real-time physics**, and an **employee database** with $\mathcal{O}(\log n)$ search using **custom BSTs**.
- Built a **Linux process monitor** by parsing **/proc** to monitor process info, and **implemented multi-level page table** simulation with 2 fork strategies, demonstrating **50% memory savings** using **copy-on-write**.

SKILLS

C, C++, Python, QML, JavaScript, Kotlin, OpenGL, WxWidgets, Qt, AWS S3, AWS Lambda, React, Node.js, Tailwind CSS, SQL, Next.js, Flask, HTML, Git, SciKit-Learn, PyTorch, Terraform, Docker, Streamlit, TensorFlow, Browser Extension Development, Data Structures & Algorithms, Object Oriented Programming, Machine Learning